

HEART DISEASE PREDICTION USING MACHINE LEARNING

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ABSTRACT

This project demonstrates how machine learning can be used to predict the presence of heart disease using patient medical data. Multiple classification models including Logistic Regression, Decision Tree, Random Forest, and Neural Network were trained and evaluated. Data preprocessing, feature scaling, model evaluation using confusion matrix and ROC curve were performed. The results show that machine learning can accurately assist in early heart disease detection.

1. INTRODUCTION

Heart disease is one of the leading causes of death worldwide. Early detection of heart disease can help prevent severe health complications and save lives. Traditional methods require time consuming medical tests and expert diagnosis. Machine learning provides an efficient way to analyse medical data and predict heart disease with high accuracy. This project focuses on building a machine learning based system to predict heart disease using patient health parameters.

2. DATASET DESCRIPTION

The dataset used in this project contains medical records of patients with multiple health related attributes.

The dataset includes the following features:

Age, Sex, Chest Pain Type, Resting Blood Pressure, Cholesterol Level, Fasting Blood Sugar, Resting ECG, Maximum Heart Rate, Exercise Induced Angina, Oldpeak, Slope, Number of Major Vessels, Thalassemia

The target column represents whether the patient has heart disease or not.
The dataset contains no missing values and was suitable for direct model training after preprocessing.

3. METHODOLOGY

The project followed a structured machine learning pipeline:

First, the dataset was loaded using the Pandas library.
Then, the data was checked for missing values and incorrect entries.
The feature columns and target column were separated.
The dataset was split into training and testing sets.
Feature scaling was applied using StandardScaler where required.

The following models were trained and tested:

Logistic Regression
Decision Tree
Random Forest
Neural Network

Each model was evaluated using accuracy, confusion matrix, and ROC curve analysis.

4. MODEL EVALUATION

Model performance was evaluated using the following techniques:

Accuracy
Accuracy is the percentage of correctly predicted values.

Confusion Matrix
The confusion matrix visualizes correct and incorrect predictions and helps analyze false positives and false negatives.

ROC Curve
The ROC curve shows how well the model separates positive and negative classes.

AUC
AUC represents the area under the ROC curve and measures overall classification performance.

5. RESULTS COMPARISON

The performance of the models was compared using accuracy scores.

Logistic Regression
Decision Tree
Random Forest
Neural Network

Among all models, Logistic Regression and Neural Network showed the best overall performance.

6. CONCLUSION

This project successfully demonstrates how machine learning can be applied to predict heart disease using patient medical data. The dataset was carefully analyzed, cleaned, and verified to contain no missing values. Proper preprocessing steps including feature target separation, train test splitting, and feature scaling were applied before training the models.

Four classification models were trained and evaluated which include Logistic Regression, Decision Tree, Random Forest, and Neural Network. Among these, Logistic Regression and Neural Network showed the strongest overall performance based on accuracy, precision, recall, F1 score, and ROC AUC values. The Random Forest model also performed well and provided stable predictions, while the Decision Tree showed comparatively lower accuracy due to its tendency to overfit.

The confusion matrix and ROC curve demonstrated that the models were able to distinguish between heart disease and non heart disease patients effectively. A high AUC value confirms that the Logistic Regression model has strong classification ability.

This project proves that machine learning can act as a reliable support tool for early heart disease detection and can be extremely useful in hospitals and preventive healthcare systems. With further optimization and real world validation, such systems can help doctors take faster and more accurate medical decisions.

7. FEATURE IMPORTANCE EXPLANATION

Feature importance represents how strongly each medical feature contributes to the prediction of heart disease. In this dataset, important contributing features include age, chest pain type, maximum heart rate achieved, oldpeak, slope, number of major vessels, and thalassemia condition. These features play a major role in determining whether a patient is likely to have heart disease.

In tree based models such as Decision Tree and Random Forest, feature importance is calculated based on how well a feature helps split the data and reduce impurity. In Logistic Regression, feature influence is determined using model coefficients, which show how strongly each feature affects the final prediction probability.

Understanding feature importance improves the transparency of the model and helps medical professionals trust the system by knowing which health indicators impact the prediction the most. This also adds strong real world relevance to the project.

8. FUTURE SCOPE

This project can be extended by using real hospital patient data.
A mobile application can be developed for live prediction.
Advanced deep learning models can be explored for further improvement.

9. REFERENCES

UCI Machine Learning Repository
Scikit Learn Documentation